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Research Paper

Evaluation of a Coinoculation Method for *Salmonella enterica* and *Enterococcus faecium* NRRL B-2354 on Whole and Broken Almonds Exposed to Dry Heat

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ABSTRACT

Traditional laboratory-based pathogen-surrogate comparison studies often involve inoculating each organism onto separate samples, which can subsequently introduce variability due to differences in food microenvironments and processing conditions. Coinoculating pathogens and surrogates onto the same sample minimizes this variability by subjecting both microorganisms to identical conditions. This study evaluated a coinoculation method for comparing the thermal resistance of *Salmonella enterica* and *Enterococcus faecium* on whole and broken almonds exposed to 168 °C for up to 12 min. Selective media were validated for accurate differentiation and enumeration of coinoculated wild-type or rifampin-resistant *Salmonella* and *E. faecium* on almonds. The media reliably distinguished and quantified each organism, with no cross-recovery observed. Reductions in *Salmonella* were not significantly different between wild-type and rifampin-resistant strains, nor between CHROMagar Salmonella and tryptic soy agar supplemented with rifampin at 50 μg/mL ($P > 0.05$). *E. faecium* demonstrated similar or greater thermal resistance compared to coinoculated *Salmonella*, supporting its use as a surrogate during dry heat treatment of almonds. While no significant difference in *Salmonella* reduction was observed between whole and broken almonds, greater reductions of *E. faecium* were observed on broken almonds, suggesting that almond structure may influence microbial thermal responses in a species-dependent manner. The findings support the utility of coinoculation for laboratory-based pathogen-surrogate comparison studies.

Low-moisture foods have been implicated in multiple foodborne outbreaks and recalls, often associated with *Salmonella enterica* (Chen et al., 2024; Harris et al., 2024; Yada & Harris, 2024). The Food Safety Modernization Act (FSMA) Preventive Control for Human Food rule requires food manufacturers, including nut processors, to validate processes used to control pathogens (U.S. Food and Drug Administration, 2018). Process validation plays a key role in ensuring the microbiological safety of low-moisture foods (Anderson et al., 2017; Dhowlaghar & Zhu, 2021; Niebuhr et al., 2008). For example, almond processors must demonstrate that they can achieve a minimum 4-log reduction of *Salmonella* for almonds grown in California and sold in North America (Federal Register, 2007). Some almond processes, such as blanching and oil roasting, may be validated by demonstrating compliance with specific time and temperature parameters (Almond Board of California [ABC], 2007a, 2007c). However, other processes such as dry roasting typically require inoculated pack or challenge studies using a surrogate organism to assess the pathogen control step under commercial conditions (ABC, 2007b, 2014; National Advisory Committee on Microbiological Criteria for Foods [NACMCF], 2010).

Selection of an appropriate surrogate organism is a critical step for process validations and challenge studies. The surrogate needs to be nonpathogenic and show similar or greater resistance to the target pathogen under the conditions being evaluated (Ceylan et al., 2021; Dhowlaghar & Zhu, 2021; Hildebrandt & Marks, 2024; Hu & Gurtler, 2017). Before directly conducting in-plant process validation studies using surrogate organisms, laboratory-based studies are usually performed to confirm and demonstrate that the selected surrogate organism shows similar survival characteristics or greater resistance than the target pathogen under specific processing conditions (Dhowlaghar & Zhu, 2021; Hu & Gurtler, 2017; Jeong et al., 2011; Niebuhr et al., 2008).

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Enterococcus faecium NRRL B-2354 has been widely recognized as a safe and conservative surrogate for *Salmonella enterica* in thermal process validation studies (Ahmad et al., 2022; ABC, 2014; Anderson et al., 2017; Brar & Danyluk, 2019; Ceylan et al., 2021; Hildebrandt & Marks, 2024; Kopit et al., 2014; Liu et al., 2018; Moussavi et al., 2020; Tsai et al., 2019; Xie et al., 2022). However, in laboratory-based pathogen-surrogate comparison studies, *Salmonella* and *E. faecium* have usually been inoculated separately into the low-moisture food(s) under study, subjected to thermal treatments in parallel, and compared posttreatment (Ahmad et al., 2022; Brar & Danyluk, 2019; Hildebrandt & Marks, 2024; Jeong et al., 2017; Moussavi et al., 2020). Separate inoculation can introduce experimental variability among samples (e.g., differences in drying rates, heat penetration, and sample handling). Such variability in laboratory-based studies can affect the consistency and accuracy of pathogen-surrogate comparisons, potentially influencing the reliability of subsequent pilot-scale or in-plant process validation studies.

Coinoculation of both a pathogen and its target surrogate organism onto the same sample matrix ensures that both microorganisms are exposed to identical food microenvironments and processing conditions, enabling a more accurate one-to-one comparison of survival kinetics and potentially improving the precision of the laboratory studies. The sensitivity of *Salmonella* and potential surrogates to propylene oxide treatment was compared using such an approach (Saunders et al., 2018). Cashew and macadamia kernels were coinoculated with a five-strain wild-type *Salmonella enterica* cocktail and single-strain surrogates (*E. faecium*, *Pediococcus acidilactici*, or *Staphylococcus carnosus*) followed by exposure to propylene oxide. *Salmonella* and the surrogate organisms were enumerated separately using selective media overlays and direct plating on selective media, respectively. Results indicated that *Salmonella* reductions were greater than those of *E. faecium* on both nut types following treatment. Despite these findings, the broader feasibility and accuracy of coinoculation methods for pathogen-surrogate comparisons in laboratory-based studies remain underexplored.

A key requirement for successful implementation of coinoculation is the ability to accurately and selectively enumerate each organism, particularly in nonsterile food matrices that may contain diverse background microbiota. The use of antibiotic-resistant variants offers a potential solution to avoid the interference of background microbiota and enable selective recovery in mixed culture on antibiotic-supplemented media. This approach has been successfully applied in food safety studies on microbial survival and transfer during postharvest handling and processing (Beuchat et al., 2012; Parnell et al., 2005; Uesugi & Harris, 2006; Xie et al., 2024, 2025).

The use of whole almonds in most almond process validation studies and their designation as the standard material in industry guidance (ABC, 2014) reflects their dominance in commercial products. Although broken kernels are also common in commercial processing lines, microbial inactivation and validation studies with them remain limited.

The present study aimed to evaluate the feasibility of coinoculating and recovering *Salmonella enterica* and *E. faecium* from almonds, a representative low-moisture food, and to assess the thermal inactivation of these bacteria on whole and broken almonds at 168 °C. The specific objectives of this study were to (1) validate selective media for accurate differentiation and enumeration of wild-type or rifampin-resistant strains; (2) confirm the suitability of *E. faecium* as a surrogate for *Salmonella* under coinoculation conditions; and (3) investigate the behavior of coinoculated *Salmonella* and *E. faecium* on whole and broken almonds during dry heat treatment.

Materials and methods

Almonds. Whole and broken almond kernels treated with propylene oxide (background microbiota determined to be <40 CFU/g on plate count agar) were obtained from Blue Diamond Growers (Sacramento, CA). Whole almond kernels (Supplementary Fig. S1A) were Nonpareil U.S. Extra No. 1 almonds of size 27/30 (27 to 30 almonds per ounce); broken almonds (Supplementary Fig. S1B) were Nonpareil U.S. No. 1 almond pieces (Supplementary Table S1) (ABC, 2022). Almonds were stored in sealed polyethylene bags (30.5 × 30.5 cm, Bitran; Com-Pac International, Carbondale, IL) inside a tightly sealed plastic container and held under refrigeration (4 ± 2 °C). Samples were used within 6 months of receipt. Prior to each experiment, almonds were removed from refrigerated storage and held at ambient temperature overnight before inoculation.

Bacterial cultures. The wild-type strain and a rifampin-resistant variant of *Salmonella enterica* serovar Enteritidis phage type (PT) 30 (*Salmonella* Enteritidis PT 30), and wild-type *E. faecium* NRRL B-2354 were used in the present study (Supplementary Table S2). The wild-type strain of *Salmonella* Enteritidis PT 30 (*Salmonella*-wt) was isolated from raw almonds associated with a 2000–2001 outbreak (Isaacs et al., 2005). The rifampin-resistant variant (*Salmonella*-rif) was selected from tryptic soy agar (TSA) made with increasing amounts of rifampin (Gold Biotechnology) (Parnell et al., 2005; Uesugi & Harris, 2006). In a single biological replicate, the survival of the wild-type and variant strains were compared on whole, separately inoculated almonds that were dried (3 days), packed in bags, and then held at ambient temperature (~23 °C) for 30 days. Although some population differences were observed at certain time points, the overall decline trends remained comparable between the wild-type and rifampin-resistant variants (data not shown). All isolates were stored at −80 °C in tryptic soy broth (TSB) supplemented with 15% glycerol. Unless otherwise specified, culture media were Difco brand (BD).

Inoculum preparation. Frozen cultures of *Salmonella*-wt and *E. faecium* were streaked onto TSA, and *Salmonella*-rif was streaked onto TSA supplemented with rifampin at 50 μg/mL (TSAR). After incubating at 37 ± 2 °C for 24 ± 4 h, an isolated and representative colony for each strain was picked from the streak plate and transferred twice in succession into 10 mL of TSB and incubated at 37 ± 2 °C for 24 ± 4 h. Then, 1 mL from each tube was plated onto a 150 × 15 mm agar plate of TSA (for *Salmonella*-wt and *E. faecium*) or TSAR (for *Salmonella*-rif), and plates were incubated for 24 ± 4 h. Three plates were prepared for both *Salmonella*-wt and -rif, and 10 plates were prepared for *E. faecium*. Bacterial lawns were then harvested by adding 9 mL (for *Salmonella*) or 6 mL (for *E. faecium*) of sterile 0.1% peptone per plate and then loosening cells with a sterile L-shaped spreader (Fisher Scientific). Cell suspensions for *Salmonella* (~25 mL from three plates, per strain) and *E. faecium* (~30 mL from 10 plates) were collected into sterile 50-mL Falcon tubes (Fisher Scientific) using sterile serological pipettes. The collected cell suspensions for *Salmonella*-wt, *Salmonella*-rif, and *E. faecium* were diluted to ~10 log CFU/mL by adding peptone (35 mL for *Salmonella*; 80 mL for *E. faecium*). Equal volumes (35 mL per strain; 70 mL for each mixture) of *Salmonella*-wt and *E. faecium* or *Salmonella*-rif and *E. faecium* were combined into a sterile specimen container and thoroughly mixed for coinoculation.

Almond coinoculation or inoculation. Briefly, whole or broken almonds were separately inoculated with the mixed inoculum preparations of *Salmonella*-wt and *E. faecium* or *Salmonella*-rif and *E. faecium* at a ratio of 1:16 (30 mL of mixed inoculum per 480 g of almonds) to obtain a final concentration of ~8 log CFU/g after inoculation (Danyluk et al., 2005). Prepared inoculum was added to the whole or broken almonds in polyethylene zipper bags (30 × 30 cm), and the almonds were alternately massaged and shaken for 2 min to ensure the uniform distribution of the inoculum. Inoculated almonds were spread in a single layer onto two sheets of filter paper (46 × 57 cm; Qualitative P5, Fisher Scientific) that had been folded in half (resulting in four layers of filter paper) and placed on a metal drying rack within a lidded plastic container. The plastic containers had been sanitized with 70% ethanol 1 day before and dried overnight. Inoculated whole

and broken almonds were allowed to dry for 3 days under ambient conditions (23 ± 1.0 °C, 35% ± 8.8% relative humidity [RH]) in the plastic containers. Populations of *Salmonella* and *E. faecium* on the almonds were measured, as described below, immediately after inoculation or coinoculation and again after drying (before thermal treatment).

Moisture and water activity. Moisture content and water activity of uninoculated and *E. faecium*-inoculated almonds were measured. Almonds (40 g) were homogenized in a commercial food processor (Waring Products) for 20 s and then sifted through a #12 testing sieve (1.7 mm; VWR Scientific). The sifted, homogenized almonds were collected in a sample pan and loaded into the moisture analyzer (HG63, Mettler-Toledo) to determine percent moisture. The water activity of the homogenized almonds was measured with a water activity meter (AquaLab 4TE, METER Group) following the manufacturer's instructions.

Validation of selective media for separate enumeration of coinoculated bacteria. To ensure the feasibility and reliability of recovering target strains only from the desired selective media, individual strains and the combined inoculum were appropriately diluted in 0.1% peptone, plated onto TSA (for all individual strains), TSAR (for *Salmonella*-rif), CHROMagar Salmonella (ChromSal; for *Salmonella*-wt and *Salmonella*-rif), and Slanetz and Bartley agar (S + B; for *E. faecium*), and incubated at 37 °C for 24 h (TSA, TSAR, ChromSal) or for 48 h (S + B).

Dry heat treatment of coinoculated almonds. Coinoculated whole or broken almonds (20 g) were spread loosely in a single layer on aluminum mesh trays (10 × 12 cm; Brite Aluminum Insect Screening; Fig. 1) at ambient temperature. A gravity flow oven (725 G, Isotemp, Fisher Scientific), fitted with a stainless-steel insert (25 × 22 cm) (Supplemental Fig. S2) that limited the loss of hot air when the oven door was open, was preheated to 168 °C. Four trays, each containing one type of coinoculated almond sample, were placed on the oven rack for heat treatment at the same time (Fig. 1): 1) whole almonds coinoculated with *Salmonella*-wt and *E. faecium*; 2) whole almonds coinoculated with *Salmonella*-rif and *E. faecium*; 3) broken almonds coinoculated with *Salmonella*-wt and *E. faecium*; or 4) broken almonds coinoculated with *Salmonella*-rif and *E. faecium*. Oven temperature was monitored using four type K thermocouples (OMEGA) secured next to the sample trays and connected to a data logger (OM-HL-EH-TC, OMEGA; Fig. 1). The oven was preheated for 30 min until all four thermocouples consistently read above 167.5 °C, with at least two readings exceeding 168 °C (marked as 168 °C; variation attributed to tray position). When samples were placed into the oven, the door remained open for ~30 s, causing a temporary temperature drop of approximately 30–40 °C; the oven returned to 168 °C within ~4 min. After trays were removed, the oven was allowed to reequilibrate for ~5 min before loading the next set of samples. Samples were collected for bacterial recovery and enumeration at 0 min (before treatment), and after holding in the oven for 4, 8, 10, or 12 min, respectively. Each experiment was repeated in four technical replicates for each sampling time, with tray positions (or sample types) rotated each time to reduce potential position bias from the observed 3 to 4 °C temperature difference across the oven (data not shown).

Recovery and enumeration. Almonds (20 g) were mixed with 40 mL of peptone (uninoculated almonds) or TSB (inoculated almonds) in 710-mL (24-oz) Whirl-Pak bags and pummeled for 2 min in a bag blender (Smasher, bioMérieux). Samples were diluted as appropriate in 0.1% peptone. Uninoculated samples were plated (250 μL) onto quadruplicate plate count agar plates, while inoculated samples were spiral-plated (50 μL per plate, in duplicate) on ChromSal (for *Salmonella*) and/or TSAR (for *Salmonella*-rif), and S + B (for *E. faecium*). Plates were incubated at 37 °C for 24 h (ChromSal, TSAR, plate count agar) or 48 h (S + B).

For the heat-treated samples, 20 g (per sample tray) were immediately transferred after oven treatment into Whirl-Pak bags with 40 mL of prechilled TSB (4 °C) to rapidly cool down the samples. Sample bags were pummeled, held at 4 °C for 5 min, diluted with 0.1% peptone as needed, and spiral-plated as described above.

Statistical analysis. Selective media validation experiments were conducted with three independent biological replicates ($n = 3$). Populations from single-strain and mixed inoculum on the same selective media were compared using Welch's *t*-test (TSAR) or one-way ANOVA (ChromSal and S + B). Heat treatments were performed in two independent biological replicates, each with four technical replicates ($n = 8$ at each treatment time). Bacterial counts below the limit of detection (LOD; 1.3 log CFU/g, based on a single colony on duplicate 50-μL plates at the lowest dilution) were excluded from mean and standard deviation calculations (Garces-Vega & Marks, 2014). At each time point, log populations from the same mixed inoculum were compared across strains (*Salmonella* versus *E. faecium*), plate types (TSAR versus ChromSal for *Salmonella*-rif), and almond types (whole versus broken) using ANOVA followed by Tukey's tests. Comparison between *Salmonella*-wt and *Salmonella*-rif on the same almond type were evaluated using a *t*-test. Data were analyzed using Excel GraphPad Prism (version 10, GraphPad Software, Inc.) and JMP 16 software (SAS Institute Inc.).

Differences in log reductions between *Salmonella*-wt and *Salmonella*-rif, ChromSal and TSAR agars, *Salmonella* and *E. faecium*, and whole and broken almonds were also evaluated by plotting paired and linear regression. Comparisons between ChromSal and TSAR agars, and between *Salmonella* and *E. faecium*, were paired within

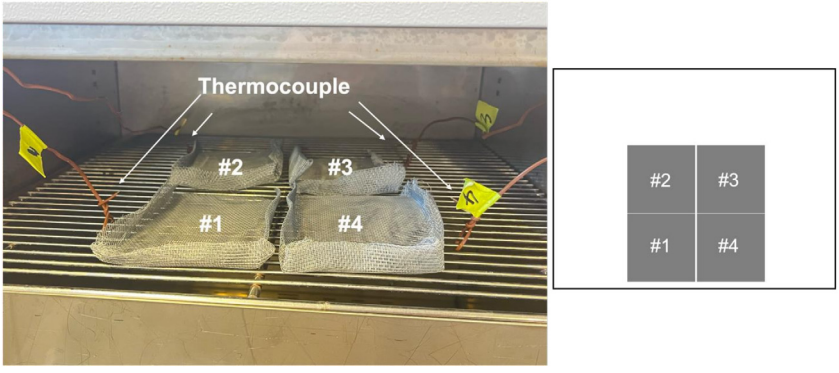


Figure 1. Arrangement of sample trays in a gravity oven during dry heat treatment of almonds on the trays. Oven temperature was monitored using four type K thermocouples (OMEGA) secured next to the sample trays and connected to a data logger (OM-HL-EH-TC, OMEGA). Each experiment was repeated in four technical replicates for each sampling time, with tray positions (or sample types) rotated each time to reduce potential position bias from the observed 3 to 4 °C temperature difference across the oven.

the same oven-treated sample. In contrast, comparisons between *Salmonella*-wt and *Salmonella*-rif, and between whole and broken almonds, were paired across separate samples that were treated concurrently in the oven. To assess overall equivalence in response across the heat treatment, linear regressions through the origin (0,0) were generated and compared to the ideal 1:1 correlation line (y = x). A perfect 1:1 correlation (slope = 1) would indicate that the paired groups have equivalent resistance and variability under the given treatment conditions (Hildebrandt & Marks, 2024). Equivalence was formally evaluated by assessing whether the 95% confidence intervals (CIs) for the slope included 1 (no significant difference) and by conducting a Wald *t*-test comparing the observed slope to the null hypothesis of slope = 1. Paired samples yielding plate counts below 25 CFU were excluded from regression analyses to ensure accuracy (Garces-Vega & Marks, 2014). The goodness of fit was evaluated using the regression coefficient (R^2) to assess the precision of the modeled linear regression.

Results and discussion

Validation of selective media for separate enumeration of coinoculated bacteria. A nonselective medium (TSA) supported the growth of wild-type and antibiotic-resistant variants of *Salmonella* and wild-type *E. faecium* (Table 1). In contrast, ChromSal, TSAR, and S + B agar, containing selective agents to inhibit unwanted organisms, enabled specific recovery of *Salmonella*, *Salmonella*-rif, and *E. faecium*, respectively. For the single-strain inoculum (liquid culture prepared from lawn), no colonies of *Salmonella*-wt were observed on TSAR or S + B (< 10 CFU/mL); populations enumerated on TSA and ChromSal were 10.7 ± 0.1 log CFU/mL and 10.7 ± 0.3 log CFU/mL, respectively (Table 1). For *Salmonella*-rif, no colonies were observed on S + B (< 10 CFU/mL), and populations from TSA, TSAR, and ChromSal were 10.7 ± 0.1, 10.7 ± 0.2, and 10.6 ± 0.2 log CFU/mL, respectively (Table 1). *E. faecium* was not detected on either TSAR or ChromSal (< 10 CFU/mL), and populations enumerated on TSA and S + B agar were 10.1 ± 0.1 log CFU/mL and 10.0 ± 0.3 log CFU/mL, respectively (Table 1).

For the mixed inoculum, both *Salmonella* and *E. faecium* grew well on TSA. Although the colonies differed in size between the two bacteria, selective media were needed to accurately enumerate each organism when coinoculated. The populations of *Salmonella* (*Salmonella*-wt or *Salmonella*-rif) from the mixed inoculum enumerated on ChromSal and/or TSAR did not differ significantly from those observed with the single-strain inoculum ($P > 0.05$; Table 1). Populations of *Salmonella*-rif in the single-strain inoculum or mixed inoculum of *Salmonella*-rif and *E. faecium* were 10.7 ± 0.2 and 10.6 ± 0.1 log

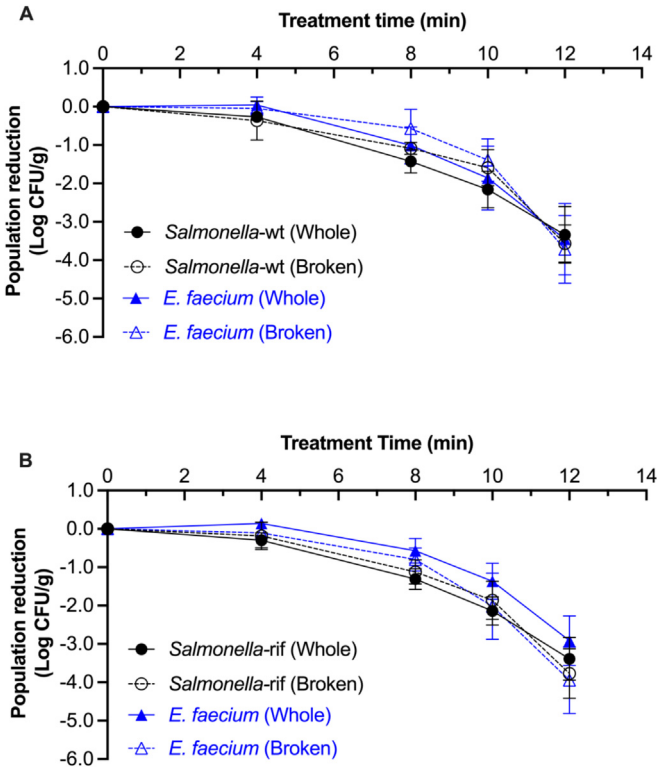


Figure 2. Reduction of coinoculated *Salmonella* Enteritidis PT 30 and *E. faecium* on whole and broken almonds during dry heat treatment in an oven with an internal temperature set to 168 °C: (A) Reduction of coinoculated parent *Salmonella* (*Salmonella*-wt) and *E. faecium*; (B) Reduction of coinoculated rifampin-resistant variant *Salmonella* (*Salmonella*-rif) and *E. faecium*; mean ± SD, $n = 8$. Almonds were coinoculated at ~7 log CFU/g (untreated, time 0). Populations of *Salmonella* and *E. faecium* were enumerated on CHROMagar Salmonella (ChromSal) and Slanetz & Bartley (S + B) agar, respectively. Populations of coinoculated *Salmonella*-wt or *Salmonella*-rif and *E. faecium* on whole and broken almonds are summarized in Supplementary Tables S3 and S4.

Table 1
Recovery and enumeration of parent *Salmonella* Enteritidis PT 30 (*Salmonella*-wt), corresponding rifampin-resistant variant (*Salmonella*-rif), and *E. faecium* inoculums on non-selective and selective media

Strain(s)	Sample type	Populations on different media (log CFU/mL or log CFU/g) [a]			
		TSA	TSAR	ChromSal	S + B
Salmonella-wt	Inoculum [b]	10.7 ± 0.1	< 1.0 [c]	10.7 ± 0.3 A	< 1.0
Salmonella-rif	Inoculum	10.7 ± 0.1	10.7 ± 0.2 A	10.6 ± 0.2 A	< 1.0
E. faecium	Inoculum	10.1 ± 0.1	< 1.0	< 1.0	10.0 ± 0.3 A
Salmonella-wt + *E. faecium*	Inoculum	ND [d]	< 1.0	10.5 ± 0.2 A	10.0 ± 0.3 A
Salmonella-rif + *E. faecium*	Inoculum	ND	10.6 ± 0.1 A	10.6 ± 0.2 A	10.0 ± 0.3 A
E. faecium	Inoculated whole almonds [e], dry heat at 168 °C for 10 min	5.8 ± 0.2	< 1.3	< 1.3	5.8 ± 0.4

[a] TSA, tryptic soy agar; TSAR, TSA supplemented with rifampin at 50 μg/mL; ChromSal, CHROMagar Salmonella; S + B, Slanetz & Bartley agar. Values are means ± standard deviation ($n = 3$). Populations from single-strain and mixed inoculum on the same selective media were compared using Welch's t-test (for TSAR) or one-way ANOVA (for ChromSal and S + B). Within each media (column), means with the same uppercase letter are not significantly different ($P > 0.05$).
[b] Liquid inoculum preparation.
[c] For liquid culture samples, the limit of detection (LOD) was 10 CFU/mL or 1 log CFU/mL. For almond samples, the LOD was 20 CFU/g (1.3 log CFU/g).
[d] ND, not determined. All strains evaluated can grow on TSA.
[e] Almonds were inoculated with *E. faecium* alone at ~7 log CFU/g before heat treatment; for enumeration after heat treatment, 10 g of almonds were added to 20 mL of tryptic soy broth (TSB), and 50 μL of the corresponding homogenate was spiral plated on media in duplicate. The LOD was 20 CFU/g or 1.3 log CFU/g.

CFU/mL, respectively when enumerated on TSAR. Populations of *E. faecium* in both mixed inoculum preparations and single-strain inoculum were 10.0 ± 0.3 log CFU/mL on S + B agar; no significant difference was observed ($P > 0.05$, Table 1). No cross-recovery was observed in the mixed inoculum: *Salmonella* was not detected (<10 CFU/mL) on S + B agar and *E. faecium* was not detected on ChromSal or TSAR. Thus, the selective media were effective in differentiating the two organisms from the mixed inoculum, indicating the reliability of selective enumeration for the coinoculation method.

Survival curves of coinoculated *Salmonella* and *E. faecium* during dry heat treatment. The survival curves for coinoculated *Salmonella* and *E. faecium* exhibited a shoulder pattern (concave down), where populations were initially stable followed by a decline (Fig. 2). During the first 4 min of the heat treatment, small (0.2 log) but significantly greater ($P < 0.05$) reductions of *Salmonella* compared to *E. faecium* (stable, 0 log) were observed (Fig. 2A and B, Supplementary Table S3). Approximately 3-log reductions were observed after exposure to 168 °C for 12 min for coinoculated *Salmonella* (*Salmonella*-wt or *Salmonella*-rif) and *E. faecium* on both whole and broken almonds (Fig. 2, Supplementary Tables S3 and S4). Reductions of coinoculated *Salmonella* and *E. faecium* were not significantly different ($P > 0.05$; Supplementary Tables S3 and S4). The data points generated from these die-off curves (Fig. 2) were subsequently used to assess the impact of rifampin resistance, plating media, and almond form.

Comparison of *Salmonella*-wt and *Salmonella*-rif reductions. Log reductions for paired samples of whole or broken almonds that were inoculated with either *Salmonella*-wt or *Salmonella*-rif and treated in the oven at the same time were compared (Fig. 3A, B). No significant differences were observed in the reductions of *Salmonella*-wt and *Salmonella*-rif when coinoculated with *E. faecium* (Fig. 3A, B). For whole almonds, the slope of the linear regression for reduction of *Salmonella*-rif as a function of the reduction of *Salmonella*-wt was 0.945 (Fig. 3A, $R^2 = 0.938$, 95% CI 0.852–1.04); not significantly different compared to the ideal regression (slope = 1). Similarly, for broken almonds, the correlation between *Salmonella*-wt and *Salmonella*-rif reductions showed a strong fit to a linear regression model (y = 1.07x, $R^2 = 0.923$; Fig. 3B), where the slope (95% CI 0.950–1.18) was not significantly different from 1. The findings support the use of

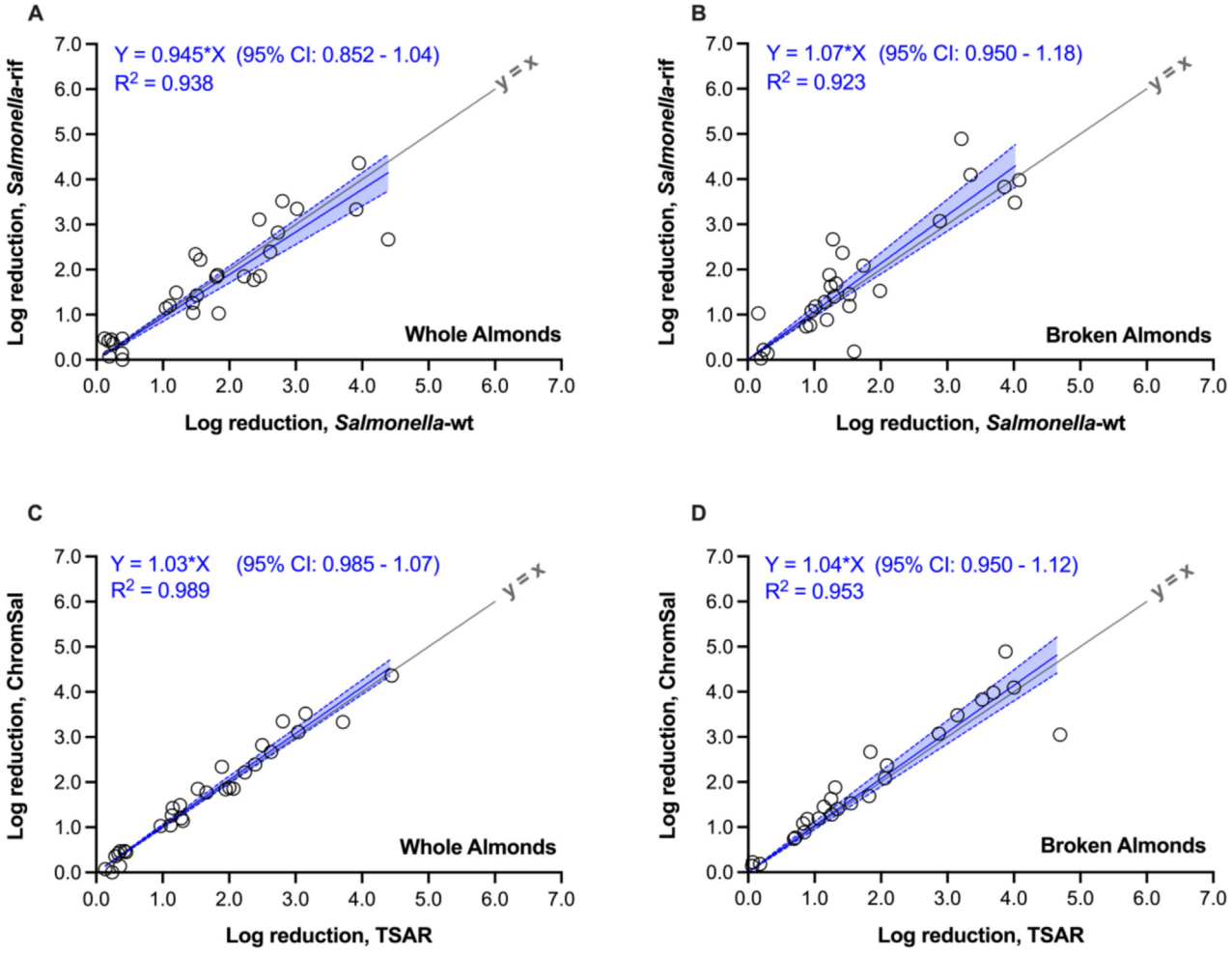


Figure 3. Log reductions of *Salmonella*-wt (A, B) and *Salmonella*-rif (A, B, C, D) during dry heat treatment of whole (A, C) and broken (B, D) almonds in an oven with an internal temperature set to 168 °C. Almonds were coinoculated with *E. faecium* at ~7 log CFU/g (untreated, time 0). Paired populations of *Salmonella*-wt and *Salmonella*-rif from the samples treated in the oven at the same time were enumerated on ChromSal (A, B); paired populations of *Salmonella*-rif from the same sample were enumerated on both ChromSal and tryptic soy agar supplemented with rifampin at 50 μg/mL (TSAR) (C, D). Solid grey lines represent the simple linear regression where y = x. Solid dark (blue) lines represent the linear regressions of *Salmonella*-rif as a function of *Salmonella*-wt (A, B), or log reductions of *Salmonella*-rif enumerated on ChromSal as a function of the log reductions on TSAR (C, D). Dashed lines and the shaded bands represent the 95% confidence interval (CI) of the fitted linear regressions. Equivalence for the paired populations was formally evaluated by assessing whether the 95% CIs for the slope included 1 (no significant difference) and by conducting a Wald *t*-test comparing the observed slope to the null hypothesis of slope = 1. Data are summarized in Supplementary Tables S3 and S4; $n = 32$.

rifampin-resistant *Salmonella* as a substitute for the wild-type strain in coinoculations for thermal validation studies.

Comparison of media for enumerating *E. faecium* and coinoculated *Salmonella*-rif after thermal treatment. Thermal treatment is a physical stressor that can cause bacterial injury through membrane damage, DNA degradation, and protein denaturation (Wesche et al., 2009). Injured cells may lose critical metabolic functions required for growth on selective media or fail to exhibit the expected color reaction, potentially leading to underestimation or false-negative results. No significant difference was observed in the recovery of *E. faecium* between TSA and S + B agar after heat treatment of whole almonds for 10 min (Table 1), suggesting minimal heat-induced injury under tested conditions. Subsequent analyses of *E. faecium* reductions were conducted using plate counts obtained from S + B agar.

Paired log reductions for the same sample were compared for coinoculated *Salmonella*-rif enumerated on TSAR and ChromSal following thermal treatment (Fig. 3C, D). Linear regressions of *Salmonella*-rif reductions enumerated using ChromSal as a function of those enumerated on TSAR yielded slopes that were not significantly different from 1 (whole almonds: 1.03 [95% CI 0.985–1.07], Fig. 3C; broken almonds: 1.04 [95% CI 0.950–1.12], Fig. 3D), with R^2 values exceeding 0.95. No significant difference in recovery was observed between TSAR and ChromSal after heat treatment. Based on these results, subsequent analyses of *Salmonella*-rif reductions were conducted using plate counts obtained from ChromSal.

Evaluation of coinoculated *E. faecium* as a surrogate for *Salmonella*-rif during dry heat treatment. To assess *E. faecium* as a surrogate, reductions of coinoculated *Salmonella*-rif and *E. faecium* within the same sample of whole or broken almonds were compared (Fig. 4A, B). For whole almonds, the reductions of *E. faecium* as a function of *Salmonella*-rif reductions fit a linear regression with a slope of 0.733 (95% CI 0.645–0.823, R^2 = 0.906, Fig. 4A), which is significantly less than 1 (P < 0.0001). The 95% CI bands fell below the ideal 1:1 correlation line, also indicating *Salmonella*-rif populations declined more rapidly than *E. faecium* under these conditions. By contrast, for broken almonds, the slope of the linear regression was 1.02 (95% CI 0.930–1.11, R^2 = 0.948, Fig. 4B); not significantly different from the ideal 1:1 relationship. This suggests that reductions of coinocu-

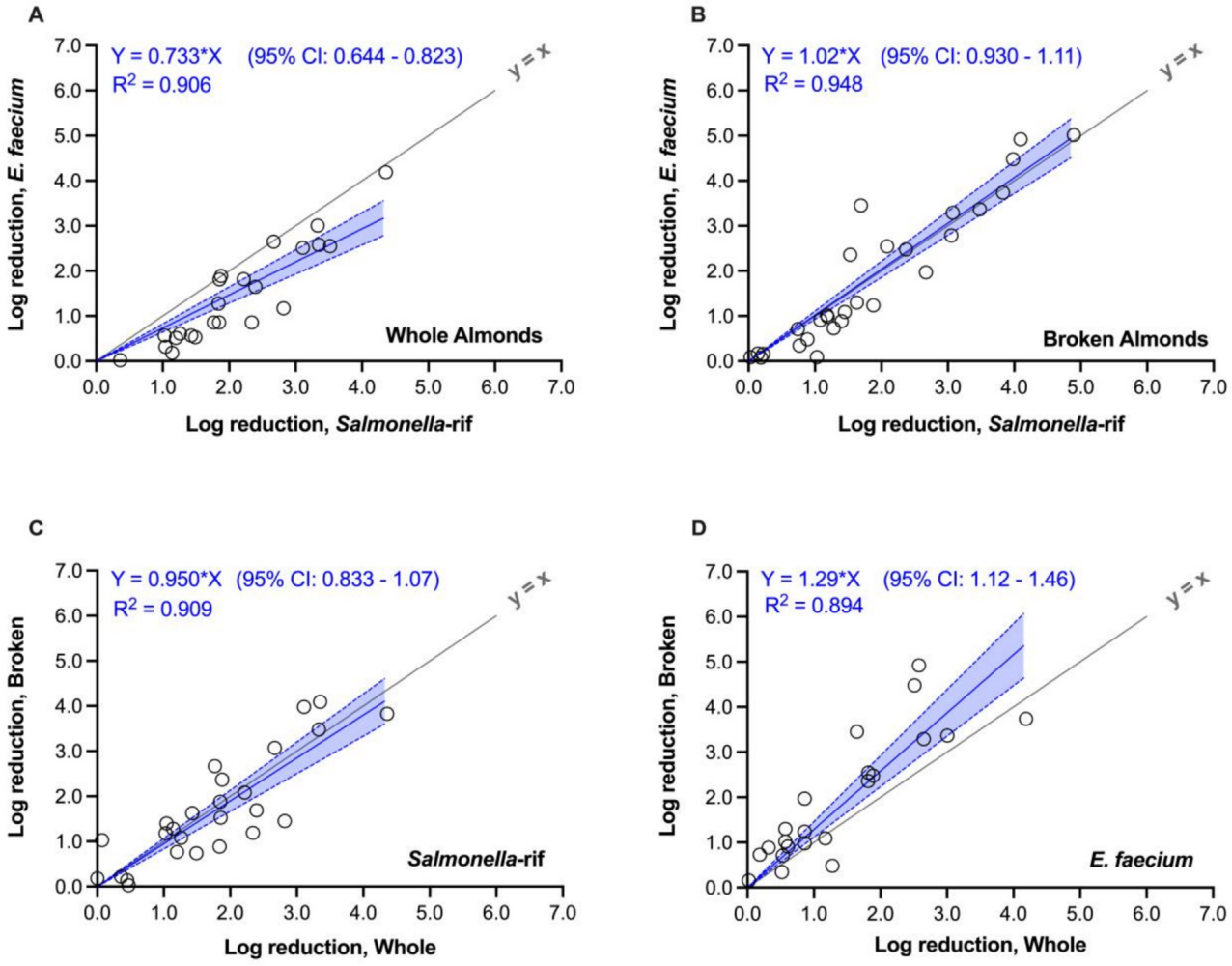


Figure 4. Log reductions of coinoculated *Salmonella*-rif (A, B, C) and *E. faecium* (A, B, D) on whole (A, C, D) and broken (B, C, D) almonds in an oven with an internal temperature set to 168 °C. Almonds were coinoculated at ~7 log CFU/g (untreated, time 0). Populations of *Salmonella*-rif and *E. faecium* were enumerated on ChromSal and S + B selective agar, respectively. Solid grey lines represent the simple linear regression where y = x. Solid dark (blue) lines represent the linear regressions of paired log reductions of *E. faecium* as a function of *Salmonella*-rif from the same sample (A, B), or paired log reductions on broken almonds as a function of the log reductions on whole almonds treated in the oven at the same time (C, D). Dashed lines and the shaded bands represent the 95% confidence interval (CI) of the fitted linear regressions. Equivalence for the paired populations was formally evaluated by assessing whether the 95% CIs for the slope included 1 (no significant difference) and by conducting a Wald *t*-test comparing the observed slope to the null hypothesis of slope = 1. Data are summarized in Supplementary Table S4; n = 32.

lated *Salmonella*-rif and *E. faecium* were equivalent on broken almonds during dry heat treatment at 168 °C. These observations of equivalent or greater thermal resistance of *E. faecium* compared with *Salmonella* are consistent with other laboratory-based studies on low-moisture foods using separately inoculated samples (Dhowlaghar & Zhu, 2021; Jeong et al., 2011), supporting the use of *E. faecium* as a surrogate for *Salmonella* during dry heating of both whole and broken almonds.

Population reductions on whole and broken almonds. Log reductions for paired whole and broken almond samples treated concurrently were plotted to evaluate the impact of almond type on microbial inactivation (Fig. 4C, D). The reductions of *E. faecium*, but not those of coinoculated *Salmonella*-rif, were influenced by whether almonds were whole or broken. For *Salmonella*-rif, log reductions on whole and broken almonds fit a linear regression with a slope of 0.950 (95% CI 0.833–1.07, $R^2 = 0.909$, Fig. 4C), indicating no significant difference between the two almond forms. In contrast, the regression slope for *E. faecium* was 1.29 (95% CI 1.12–1.46), with the 95% CI bands above the ideal 1:1 correlation (Fig. 4D), suggesting that *E. faecium* population declines were significantly greater on broken almonds compared with whole almonds under the same thermal treatment ($P = 0.0017$).

The differing behaviors of *Salmonella* and *E. faecium* might be attributed to structural differences between gram-negative (*Salmonella*) and gram-positive (*E. faecium*) bacteria, as well matrix-related factors associated with whole versus broken almonds. Broken almonds have greater surface area and exposed internal surfaces, which may lead to more rapid moisture loss and increased oil release. These physical changes may influence come-up time and heat penetration during dry heat treatment. Moisture loss can initially buffer heating through evaporative cooling, but once the surface dries, the now drier matrix heats up more rapidly due to its lower specific heat capacity (Fellows, 2009). Oil release may reduce convective heat transfer due to the lower thermal conductivity of lipids compared with water. Previous studies have shown that oil can enhance the thermal resistance of *Salmonella* in oil-rich or high-fat low-moisture foods (Deng et al., 2012; Li et al., 2014; Liu et al., 2023; Yang et al., 2020a, 2020b).

At the cellular level, thermal exposure and desiccation impose osmotic stressors that can lead to alterations in bacterial membrane composition, including changes to fatty acid and phospholipid profiles (Chen et al., 2014). In gram-positive organisms like *E. faecium*, which lack an outer membrane, direct interactions with oil may disrupt cytoplasmic membrane integrity thereby increasing susceptibility to heat. In contrast, the outer membrane of gram-negative bacteria like *Salmonella*, which is rich in lipopolysaccharides, may confer additional protection against thermal and desiccation stresses. Recent studies have shown that *Salmonella* cells in peanut oil maintain low levels of transcriptional activity; however, key stress-response genes such as heat shock proteins, remain active, potentially providing cross-protection against environmental stressors (Deng et al., 2012; Liu et al., 2023). Additionally, the lipid structure of the oil itself may influence bacterial metabolic adaptation (Zhao et al., 2026). Future research should investigate whether similar lipid-mediated protective mechanisms occur in gram-positive organisms such as *E. faecium*, to further understand the role of lipids in modulating thermal resistance.

The present study used a coinoculation method to demonstrate that the thermal tolerance of *Salmonella* and *E. faecium* was influenced by almond form, with *E. faecium*, but not *Salmonella,* showing greater reductions on broken almonds compared to whole almonds. The equivalent thermal resistance on broken almonds and greater resistance on whole almonds suggests that *E. faecium* can serve as a conservative surrogate for *Salmonella* during dry heat treatment of both whole and broken almonds. However, additional research is needed to evaluate surrogate performance across a broader range of temperatures, particularly for broken almonds. In addition to providing almond-specific insights, this study also establishes a practical framework for applying coinoculation methods to other low-moisture foods, enabling controlled, side-by-side comparisons of target pathogens and surrogate organisms under identical conditions. The use of selective media along with rifampin-resistant *Salmonella* was validated for accurate differentiation and enumeration of coinoculated strains with no observed cross-recovery. Future research should assess the broader applicability of this approach across diverse low-moisture food matrices and processing conditions.

CRediT authorship contribution statement

Yucen Xie: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis. **Christopher Theofel:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Vanessa Lieberman:** Writing – review & editing, Methodology, Investigation. **Linda J. Harris:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jfp.2025.100644.

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